ThermoFisher SCIENTIFIC

# SAFETY DATA SHEET


**Creation Date** 04-Jun-2010
**Revision Date** 13-May-2024
**Version** 3

ALFAAR17510

## 2-Chloroethanol, 99+%

## SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

| | |
|---|---|
| **产品说明:** | **2-Chloroethanol, 99+%** |
| **Product Description:** | **2-Chloroethanol, 99+%** |
| **Cat No. :** | **R17510** |
| **Synonyms** | Ethylene chlorohydrin |
| **CAS No** | 107-07-3 |
| **Molecular Formula** | C2 H5 Cl O |
| **Supplier** | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608 |
| **Emergency Telephone Number** | For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11<br>Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99<br>**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887 |
| **E-mail address** | begel.sdsdesk@thermofisher.com |
| **Recommended Use** | Laboratory chemicals. |
| **Uses advised against** | No Information available |

## SECTION 2. HAZARD IDENTIFICATION

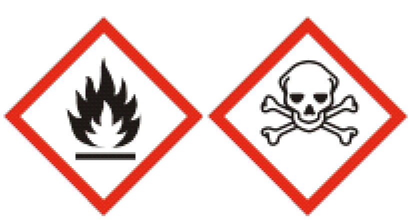
| **Physical State** | **Appearance** | **Odor** |
|---|---|---|
| Liquid | Clear | Petroleum distillates |

**Emergency Overview**
Flammable liquid and vapor. Fatal if swallowed. Fatal in contact with skin. Fatal if inhaled.

**Classification of the substance or mixture**

| | |
|---|---|
| Flammable liquids. | Category 3 |
| Acute Oral Toxicity | Category 2 |
| Acute Dermal Toxicity | Category 1 |
| Acute Inhalation Toxicity - Vapors | Category 1 |

**Label Elements**



**Signal Word** **Danger**

**Hazard Statements**
H226 - Flammable liquid and vapor
H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

**Precautionary Statements**
**Prevention**
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
P240 - Ground and bond container and receiving equipment
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment
P242 - Use non-sparking tools
P243 - Take action to prevent static discharges
P262 - Do not get in eyes, on skin, or on clothing
P264 - Wash face, hands and any exposed skin thoroughly after handling
P270 - Do not eat, drink or smoke when using this product
P271 - Use only outdoors or in a well-ventilated area
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P284 - Wear respiratory protection
**Response**
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P310 - Immediately call a POISON CENTER or doctor
P330 - Rinse mouth
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish
P362 + P364 - Take off contaminated clothing and wash it before reuse
**Storage**
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed
P405 - Store locked up
**Disposal**
P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**
Flammable liquid. Vapors may cause flash fire or explosion.
**Health Hazards**
Very toxic if swallowed. Fatal in contact with skin. Fatal if inhaled.
**Environmental hazards**
Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.  . Will likely be mobile in the environment due to its water solubility.  The product is water soluble, and may spread in water systems.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No | Weight % |
|---|---|---|
| Ethylene chlorohydrin | 107-07-3 | >95 |

## SECTION 4. FIRST AID MEASURES

**Eye Contact**
Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**
Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

**Inhalation**
Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**
Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**
Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**
Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**
Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**
Water spray, carbon dioxide ($CO_2$), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**
No information available.

**Specific Hazards Arising from the Chemical**
Flammable. Very toxic. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**
Wear self-contained breathing apparatus and protective suit. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing.

**Environmental Precautions**
Avoid release to the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**
Wear self-contained breathing apparatus and protective suit. Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

**Handling**
Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Use spark-proof tools and explosion-proof equipment. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges.

**Storage**
Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

**Specific Use(s)**
Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | China | Taiwan | Thailand | Hong Kong |
|---|---|---|---|---|
| Ethylene chlorohydrin | Skin<br>Ceiling: 2 mg/m³ | TWA: 1 ppm<br>TWA: 3.3 mg/m³ | TWA: 5 ppm | - |

| Component | ACGIH TLV | OSHA PEL | NIOSH | The United Kingdom | European Union |
|---|---|---|---|---|---|
| Ethylene chlorohydrin | Ceiling: 1 ppm<br>Skin | Skin<br>(Vacated) Ceiling: 1 ppm<br>(Vacated) Ceiling: 3 mg/m³<br>TWA: 5 ppm<br>TWA: 16 mg/m³ | IDLH: 7 ppm<br>Ceiling: 1 ppm<br>Ceiling: 3 mg/m³ | STEL: 1 ppm 15 min<br>STEL: 3.4 mg/m³ 15 min<br>Skin | |

*Legend*

*ACGIH - American Conference of Governmental Industrial Hygienists*
*OSHA - Occupational Safety and Health Administration*
*NIOSH: NIOSH - National Institute for Occupational Safety and Health*

**Monitoring methods**
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

### Exposure Controls

**Engineering Measures**
Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments |
|---|---|---|---|---|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | - | EN 374 | (minimum requirement) |

Inspect gloves before use.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.
Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use

| | |
|---|---|
| | appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly |
| **Large scale/emergency use** | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br>**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387 |
| **Small scale/Laboratory use** | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br>**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| **Hygiene Measures** | Handle in accordance with good industrial hygiene and safety practice. |
| **Environmental exposure controls** | Prevent product from entering drains. Do not allow material to contaminate ground water system. |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

| | | |
|---|---|---|
| **Appearance** | Clear | |
| **Physical State** | Liquid | |
| **Odor** | Petroleum distillates | |
| **Odor Threshold** | No data available | |
| **pH** | 6-7 | 500 g/l aq.sol |
| **Melting Point/Range** | -63 °C / -81.4 °F | |
| **Softening Point** | No data available | |
| **Boiling Point/Range** | 129 - 130 °C / 264.2 - 266 °F | |
| **Flash Point** | 55 °C / 131 °F | **Method -** No information available |
| **Evaporation Rate** | No data available | |
| **Flammability (solid,gas)** | Not applicable | Liquid |
| **Explosion Limits** | **Lower** 5 Vol%<br>**Upper** 16 Vol% | |
| **Vapor Pressure** | 6.6 mbar @ 20 °C | |
| **Vapor Density** | 2.78 (Air = 1.0) | (Air = 1.0) |
| **Specific Gravity** / **Density** | 1.200 | |
| **Bulk Density** | Not applicable | Liquid |
| **Water Solubility** | Miscible | |
| **Solubility in other solvents** | No information available | |
| **Partition Coefficient (n-octanol/water)** | | |
| **Component** | **log Pow** | |
| Ethylene chlorohydrin | 1.06 | |
| **Autoignition Temperature** | 425 °C / 797 °F | |
| **Decomposition Temperature** | No data available | |
| **Viscosity** | 3.4 mPa s at 20 °C | |
| **Explosive Properties** | | explosive air/vapour mixtures possible |
| **Oxidizing Properties** | No information available | |
| **Molecular Formula** | C2 H5 Cl O | |
| **Molecular Weight** | 80.51 | |

## SECTION 10. STABILITY AND REACTIVITY

| | |
|---|---|
| **Stability** | Moisture sensitive. |

**Hazardous Reactions** No information available.
**Hazardous Polymerization** No information available.

**Conditions to Avoid** Incompatible products. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

**Materials to avoid** Bases. Metals. Strong acids.

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide ($CO_2$). Phosgene. Hydrogen chloride gas.

## SECTION 11. TOXICOLOGICAL INFORMATION

**Product Information**

**(a) acute toxicity;**

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|---|---|---|---|
| Ethylene chlorohydrin | LD50 = 71 mg/kg ( Rat ) | LD50 = 67 mg/kg ( Rabbit ) | 0.11 mg/L/4h (37 ppm) (Rat) |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**
**Respiratory** No data available
**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Other Adverse Effects** See actual entry in RTECS for complete information

**Symptoms / effects,both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** This product contains the following substance(s) which are hazardous for the environment.

| Component | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|---|---|---|---|---|
| Ethylene chlorohydrin | LC50: 30.8 - 41.2 mg/L, | EC50: 212 mg/L/48h | EC50: 72 2 mg/L/72h | EC50 = 390.8 mg/L 15 |

| | | | | |
|---|---|---|---|---|
| | 96h flow-through<br>(Oncorhynchus mykiss)<br>LC50: 49 - 84 mg/L,<br>96h static (Pimephales<br>promelas)<br>LC50: 26.4 - 34.5 mg/L,<br>96h flow-through<br>(Oryzias latipes)<br>LC50: 19.2 - 24.1 mg/L,<br>96h flow-through<br>(Lepomis macrochirus)<br>LC50: 35 - 40 mg/L,<br>96h flow-through<br>(Pimephales promelas) | | | min<br>EC50 = 9000 mg/L 9 h<br>EC50 = 9600 mg/L 17 h |

**Persistence and Degradability** Expected to be biodegradable
**Persistence** Miscible with water, Persistence is unlikely, based on information available.
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|---|---|---|
| Ethylene chlorohydrin | 1.06 | No data available |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors
**Persistent Organic Pollutant** This product does not contain any known or suspected substance
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14. TRANSPORT INFORMATION

**Road and Rail Transport**

**UN-No** UN1135
**Proper Shipping Name** ETHYLENE CHLOROHYDRIN
**Hazard Class** 6.1
**Subsidiary Hazard Class** 3
**Packing Group** I

**IMDG/IMO**

**UN-No** UN1135

**Proper Shipping Name** ETHYLENE CHLOROHYDRIN
**Hazard Class** 6.1
**Subsidiary Hazard Class** 3
**Packing Group** I

**IATA** FORBIDDEN FOR IATA TRANSPORT

**UN-No** UN1135
**Proper Shipping Name** ETHYLENE CHLOROHYDRIN, FORBIDDEN FOR IATA TRANSPORT
**Hazard Class** 6.1
**Subsidiary Hazard Class** 3
**Packing Group** I

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**
X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ethylene chlorohydrin | X | X | X | X | 203-459-7 | X | X | X | X | X | X | KE-05650 |

**National Regulations**

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department
**Creation Date** 04-Jun-2010
**Revision Date** 13-May-2024
**Revision Summary** New emergency telephone response service provider.

**Training Advice**
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances
**IECSC** - Chinese Inventory of Existing Chemical Substances
**KECL** - Korean Existing and Evaluated Chemical Substances
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List
**ENCS** - Japanese Existing and New Chemical Substances
**AICS** - Australian Inventory of Chemical Substances
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit
**ACGIH** - American Conference of Governmental Industrial Hygienists
**TWA** - Time Weighted Average
**IARC** - International Agency for Research on Cancer

**DNEL** - Derived No Effect Level
**RPE** - Respiratory Protective Equipment
**LC50** - Lethal Concentration 50%
**NOEC** - No Observed Effect Concentration
**PBT** - Persistent, Bioaccumulative, Toxic

**PNEC** - Predicted No Effect Concentration
**LD50** - Lethal Dose 50%
**EC50** - Effective Concentration 50%
**POW** - Partition coefficient Octanol:Water
**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association
**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road
**OECD** - Organisation for Economic Co-operation and Development
**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code
**MARPOL** - International Convention for the Prevention of Pollution from Ships
**ATE** - Acute Toxicity Estimate
**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS



## End of Safety Data Sheet